

LÓRÁNT NAGY, PHD

lorant-nagy.github.io

CONTACT INFORMATION

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EDUCATION AND EMPLOYMENT

2015 - 2018	MSc, Mathematics	Budapest University of Technology and Economics
2017 - 2018	Internship	Morgan Stanley & Co LLC
2018 - 2022	PhD, Mathematics	Central European University
2019 -	Research Assistant	Alfred Renyi Institute of Mathematics
2023 -	Research Assistant	Eötvös Lóránd University – AI Research Group
2024 .	Research Assistant	Artificial Intelligence National Laboratory

PUBLICATIONS AND RESEARCH

Rásonyi, M, Nagy, L, *Optimal long-term investment in illiquid markets when prices have negative memory*, Electronic Communications in Probability. 2021; 26

Guasoni, P, Nagy, L, Rásonyi, M. *Young, timid, and risk takers* Mathematical Finance. 2021; 31: 1332–1356.

Rasonyi, M, Nagy, L, *On arbitrage with linear and superlinear friction* In preparation.

Boros D, Csanády B, Ivkovic I, Nagy L, Lukács A, Márkus L. *Deep learning the Hurst parameter of linear fractional processes and assessing its reliability* Qual Reliab Engng Int. 2024; 40: 4228–4246.

Csanády, B, Nagy, L, Boros, D, Ivkovic, I, Kovács, D, Tóth-Lakits, D, Márkus, L, Lukács, A. *Parameter Estimation of Long Memory Stochastic Processes with Deep Neural Networks* ECAI 2024. 2024; 392

Rásonyi, M, Nagy, L *Optimal investment with transient price impact and stochastic spread* arxiv at 2511.12093

EXPOSURE TO INDUSTRY, OTHER SKILLS

programming skills, fluent in Python with high exposure to PyTorch, Tensorboard, comfortable with Rust and C++, familiar with the Wolfram computer algebra system, some Docker, lot of experience with Python science libraries and visualization tools

investment banking, Morgan Stanley, testing in-house pricing systems, responsibility for the recurrent reviewing of a few low priority products, co-operative work with the interest rate modeling team

medical sector, diagnosis and mortality forecast with machine learning techniques and deep learning

power industry, ALTEO, energy market modeling and forecasting intraday prices

SEMINAR PRESENTATIONS AND CONFERENCE TALKS

- 2020; Alfréd Rényi Institute of Mathematics, Probability Seminar, *How to invest in mean-reverting markets?*
- 2020; Central European University, PhD Seminar, *Investment in mean-reverting markets*
- 2021; Alfréd Rényi Institute of Mathematics, Probability Seminar, *On sub-Gaussian estimators of the mean*
- 2022; University of Michigan, Financial Mathematics Seminar, *Young, Timid, and Risk Takers*
- 2023; Alfréd Rényi Institute of Mathematics, Probability Seminar, *Random Walks in Random Environments*
- 2023; Eötvös Loránd University, ELTE-Rényi Mi Workshop, *Risk averse trading*
- 2024; Bolyai Intézet, AMK, *Optimal execution in superlinearly mean-reverting environments*
- 2025; Technische Universität Wien, VCMF2025, *Non-convex optimal investment with transient price impact*

MISCELLANEOUS

- . 2025, Intermediate-level Qualification in Copyright Law, Hungarian Intellectual Property Office
- . 2022, Award for Advanced Doctoral Students
- . 2019, Teaching Fundamentals of Stochastic Analysis course at the Central European University
- . 2016, Second Place Award at MSc uni. competition with "Examples in the theory of continuous trading with friction"